

ADVANCED MATERIALS

Supporting Information

for *Adv. Mater.*, DOI 10.1002/adma.202304415

Compact Hole-Selective Self-Assembled Monolayers Enabled by Disassembling Micelles in Solution for Efficient Perovskite Solar Cells

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Methods

Materials

All the reagents were used as received without any further purification. Cesium iodide (CsI), methylammonium iodide (MAI), methylammonium chloride (MACl), bathocuproine (99.9%) and C60 were bought from Xi'an Polymer Light Technology. Formamidinium iodide (FAI) was purchased from Dysol. PbI_2 (99.985%), MeO-2PACz and 2PACz were obtained from TCI. dimethylformamide (DMF), dimethyl sulfoxide (DMSO), dimethylacetamide (DMAC) and N-methyl-2-pyrrolidone (NMP), isopropanol (IPA) and chlorobenzene (CB) are purchased from J&K. Compound CbzNaph (ref 24) and Piperazinium iodide (PI) (ref 9) were synthesized according our group previous report.

Device fabrication

ITO glass was cleaned by sonication with detergent (Decon 90), deionized water, acetone and IPA for 30 min. Then the washed ITO galss was dried in an 80 °C oven for over 6 h. Then the dried ITO substrates were treated with UV ozone for 30 min before being transferred into nitrogen glovebox for the device fabrication. The SAM solutions IPA-CbzNaph (1.5 mg/ml), IPA-MeO-2PACz (1mg/ml) and IPA-2PACz (1mg/ml) were prepared by dissolving the CbzNaph, MeO-2PACz and 2PACz in IPA under stirring and 60 °C. For the SAMs prepared by co-solvent, the CbzNaph, MeO-2PACz and 2PACz were first dissolved in DMF to obtain 50 mg/ml solutions. Then the solutions were diluted with IPA to obtain the target solutions Cos-CbzNaph (1.5 mg/ml, in DMF/IPA 3.3% v/v), Cos-MeO-2PACz (1mg/ml in DMF/IPA 2% v/v) and 2PACz (1mg/ml in DMF/IPA 2% v/v). Then the 40 uL of the SAM solution was dropped

onto the ITO substrates and spin-coated at 3000 rpm for 50 s. Then after annealing at 100 °C for 10 min, the substrates were cooled down to room temperature and washed with IPA through spin-coating at 3000 rpm for 30 s, then annealed at 100 °C for 5 min. Then, 1.6 M perovskite precursor solution was prepared by mixing FAI, PbI₂, MAI and CsI in DMF:DMSO mixed solvent (4:1 v:v) with the chemical formula Cs_{0.05}MA_{0.15}FA_{0.80}PbI₃. 15 mol% of MACl was added to the precursor before deposition. After stirring for over 6 h, the precursor is transferred for spin-coating. 50 uL of the precursor was dripped onto SAM-based substrates and spin-coating at 1000 rpm for 7 s and 5000 rpm for 39 s, 180μL CB as anti-solvent is dripped at the final 20 s. Then the perovskite films were annealed at 100 °C for 30 min. For the surface passivation, PI (0.3 mg/mL in IPA) was spin-coated on perovskite film at 4000 rpm for 30 s, then annealed at 100 °C for 10 min. The films were transferred to thermal evaporator for deposition of C60, BCP and Ag with a thickness of 25 nm, 6 nm and 100 nm under a high vacuum ($<4 \times 10^{-6}$ torr). For the anti-reflection coating, a MgF₂ layer with a thickness of 110 nm was thermal evaporated onto the backside of devices.

Characterization

Dynamic light scattering (DLS) experiments were conducted by a dynamic light scattering particle size Analyzer (Malven Zetasizer Nano ZS) at 25 °C with a monochromatic coherent He-Ne laser (640 nm) as the light source. An avalanche photodiode detector that detected the scattered light at an angle of 173°. DLS measurements were carried out in SAM solutions in IPA or co-solvent to determine the size of micelles and the critical micelle concentration (CMC)

of SAMs in IPA. Transmission electron microscopy (TEM) measurements were performed with a Hitachi HT7700 instrument at a voltage of 120 kV. The specimen was prepared by drop-casting the sample solution onto a carbon-coated copper grid and air-dried before measurement.

Contact angle was measured with a DataPhysics contact angle tester and the water drop volume is set as 1 μ L. Time-dependent contact angle tests can be exponentially fitted with the following equation:

$$y=y_0+A\exp(-x/k) \quad \text{Equation S1}$$

Where y_0 and A are fitting constants. k is the decay constant.

The UPS and XPS characterization were performed by a VG ESCALAB 220i-XL surface analysis system equipped with a He discharge lamp ($h\nu = 21.22$ eV) and a monochromatic Al-K α X-ray gun ($h\nu = 1486.6$ eV). The SAM solutions were deposited on ITO in the same process as device fabrication. Typically, the characterized peak of hydrocarbon C1s from adventitious carbon at 284.8 eV was used for binding energy calibration. The surface morphology of perovskite films and the energy dispersive X-ray spectroscopy (EDS) mappings were characterized by scanning electron microscopy (SEM, Philips XL30 FEG). Cyclic voltammetry (CV) measurements were conducted on a CHI660D electrochemical workstation. The CV experiments were performed at room temperature in an argon atmosphere with a three-electrode system using a bare ITO or SAM-modified ITO as the working electrode, Pt wire as the counter electrode, and an Ag/AgCl (saturated KCl) as the reference electrode. Tetrabutylammonium phosphorus hexafluoride (Bu₄NPF₆, 0.1M) in *o*-DCB solution was used as the supporting electrolyte, and a series of scan rate was applied. For calibration, the redox potential of ferrocene/ferrocenium (Fc/Fc⁺) was measured under the same condition.

The fluorescence spectra of CbzNaph, MeO-2PACz and 2PACz in IPA solution were conducted by a spectrophotometer Fluoromax-4 and the samples were excited at 340 nm. In-situ photoluminescence (in-situ PL) spectra were characterized with home-made equipment, including an excitation system, fiber system and detector system. An excitation laser (330 nm) was introduced to the sample through a fiber, and the PL spectra were detected by using a detector connected to an Ocean Optics USB2000. The 40 μ L of the SAM solution was dropped onto the ITO substrates and spin-coated at 300 rpm for 50 s, then annealing at 100 $^{\circ}$ C for 10 min. The XRD patterns for perovskite film crystallization analysis were measured by an X-ray diffraction (XRD) using a Bruker D2 Phaser with Cu K α radiation. Steady-state photoluminescence (PL) and time-resolved photoluminescence spectrum (TRPL) spectra are measured with a FLS1000 photoluminescence spectrometer system (Edinburgh) with an excitation light of 450 nm and a pulsed excitation laser of 485 nm, respectively

J-V characteristics of PSC devices were measured in nitrogen glovebox at room temperature by using a Keithley 2400 source meter under simulated sunlight from a solar simulator (Enlitech, SS-F5, Taiwan). EQE is measured by an EnLi Technology (Taiwan) EQE measurement system. A National Renewable Energy Laboratory calibrated silicon solar cell is used to calibrate the AM 1.5G light intensity. The PVCs are equipped with a shading mask with an aperture area of 0.04 cm² to ensure the accuracy of current density in J-V curves. The operational stability tests were carried out at the MPP for the unencapsulated cells under AM 1.5G illumination in N₂ atmosphere which was produced by a xenon-lamp-based solar simulator. The bias at the MPP was calculated and applied automatically. The light intensity was calibrated by a standard silicon reference cell from Newport.

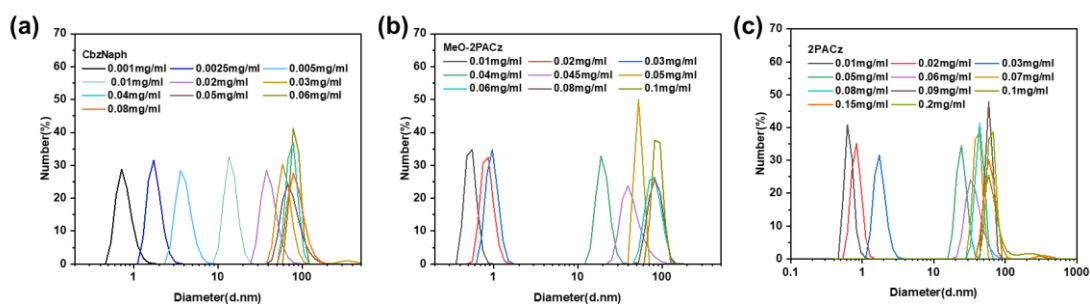


Figure S1. Size distribution determined by DLS in different concentrations of (a) CbzNaph (b) MeO-2PACz and (c) 2PACz in IPA.

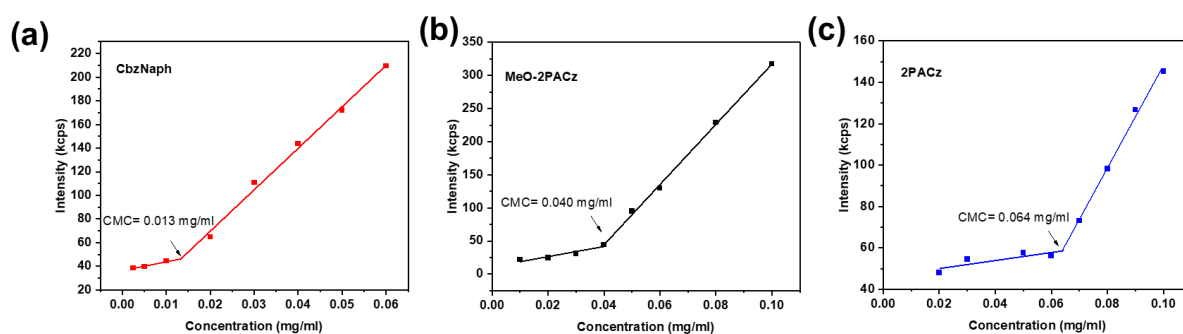


Figure S2. Light scattering intensity at different concentrations of (a) CbzNaph, (b) MeO-2PACz and (c) 2PACz in IPA

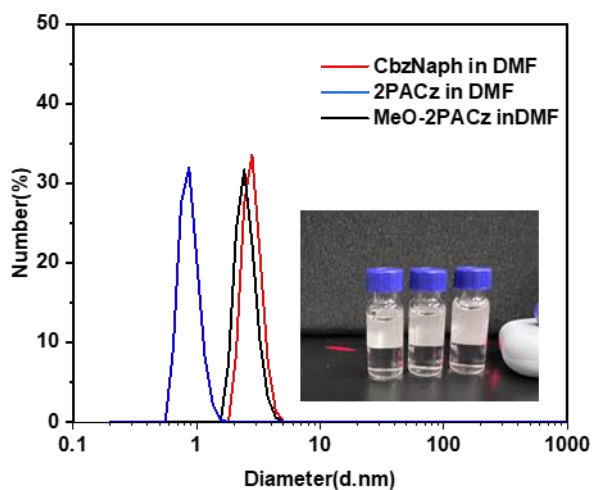


Figure S3. Size distribution determined by DLS of CbzNaph, MeO-2PACz and 2PACz in DMF, the inset picture shows that there is no Tyndall effect of these SAM solutions in DMF.

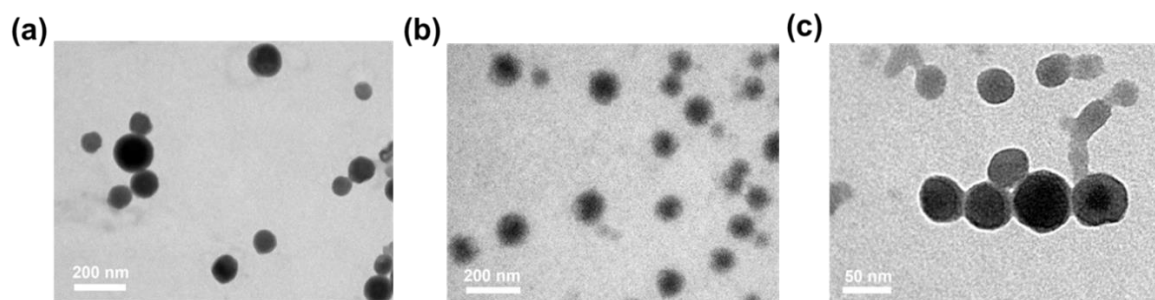


Figure S4. TEM results of micelles formed from (a) CbzNaph, (b) MeO-2PACz and (c) 2PACz in IPA.

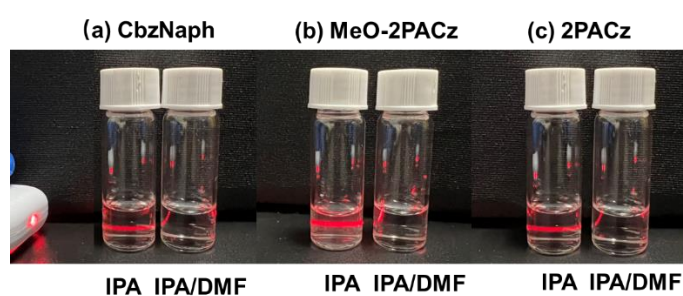


Figure S5. Light scattering effect of (a) CbzNaph, (b) MeO-2PACz and (c) 2PACz solution prepared by IPA (left) and co-solvent (right).

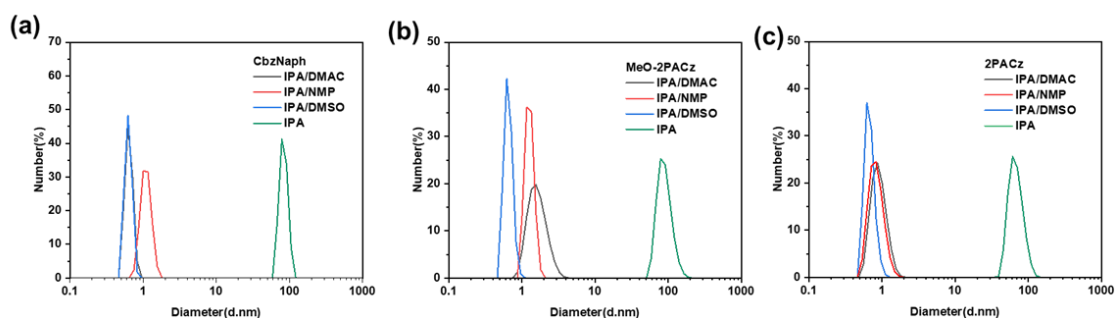


Figure S6. Size distribution determined by DLS of (a) CbzNaph (b) MeO-2PACz and (c) 2PACz in IPA and different co-solvents

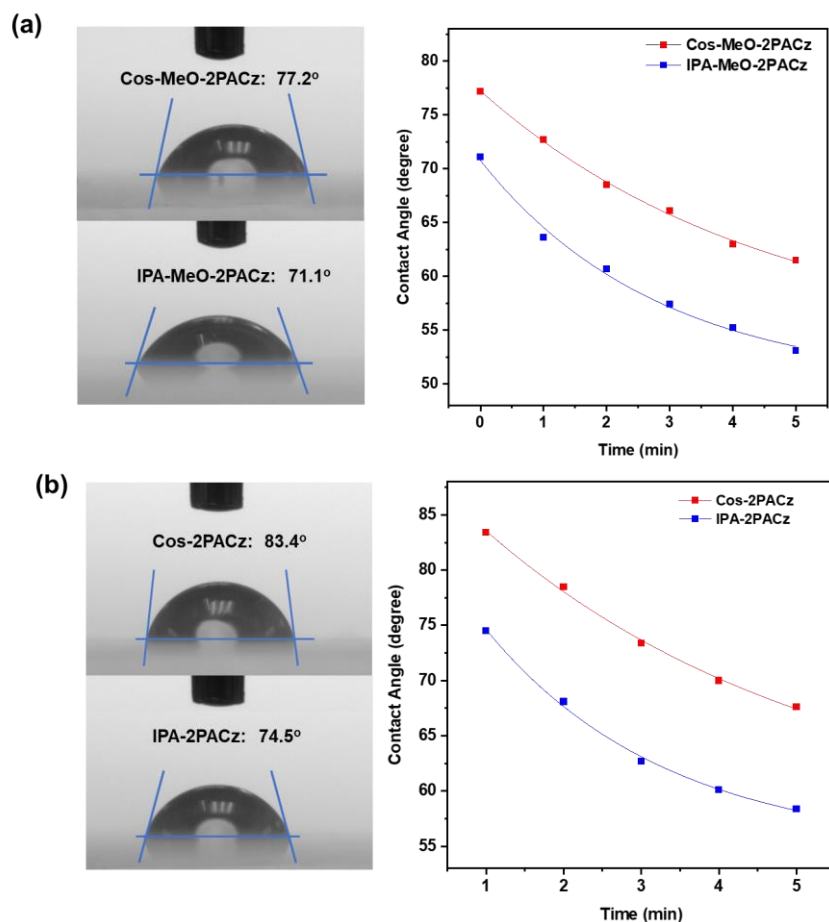


Figure S7. Water contact angles (left) and Time-dependent contact-angle test(right) of (a) Cos-MeO-2PACz and IPA-MeO-2PACz (b) Cos-2PACz and IPA-2PACz modified ITO.

Table S1. The initial contact angles and the fitting parameters for Cos-CbzNaph, IPA-CbzNaph, Cos-MeO-2PACz, IPA-MeO-2PACz, Cos-2PACz and IPA-2PACz modified ITO.

	Cos-CbzNaph	IPA-CbzNaph	Cos-MeO-2PACz	IPA-MeO-2PACz	Cos-2PACz	IPA-2PACz
Initial contact angle (°)	87.2	81.3	77.2	71.1	83.4	74.5
Decay constant k (min ⁻¹)	4.41	3.03	4.52	2.82	4.35	2.33
y ₀	64.1	59.7	53.5	49.9	56.7	54.7
A	22.9	21.2	23.7	20.8	33.8	30.7

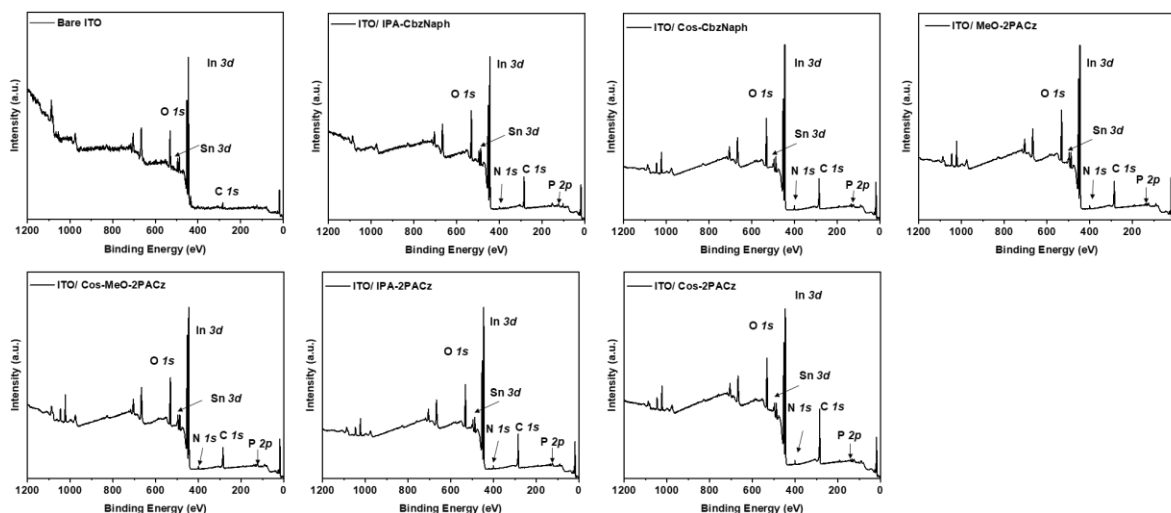


Figure S8. XPS spectra of bare ITO, IPA-CbzNaph, Cos-CbzNaph, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-2PACz and Cos-2PACz modified ITO.

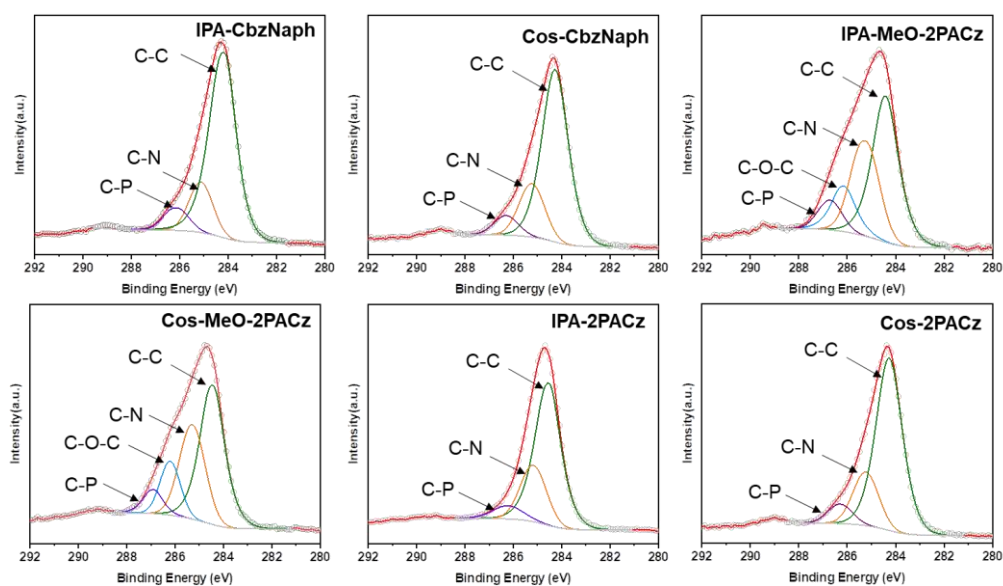


Figure S9. X-ray photoelectron spectra of C 1s for IPA-CbzNaph, Cos-CbzNaph, IPA-MeO-2PACz, Cos-MeO-PACz, IPA-2PACz and Cos-2PACz modified ITO.

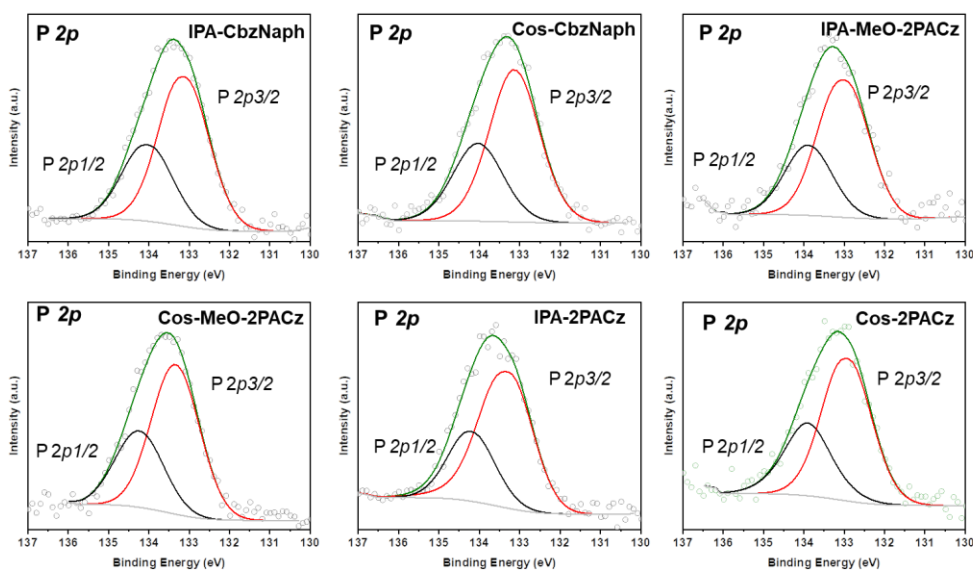


Figure S10. X-ray photoelectron spectra of P 2p for IPA-CbzNaph, Cos-CbzNaph, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-2PACz and Cos-2PACz modified ITO.

Table S2. Atomic concentration determined by XPS for IPA-CbzNaph, Cos-CbzNaph, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-2PACz and Cos-2PACz modified ITO.

	Atomic concentration (%)						
	C 1s	N 1s	O 1s	P 2p	In 3d	Sn 3d	P/In
ITO	10.60	-	59.64		26.70	3.06	-
ITO+IPA-CbzNaph	44.34	1.96	37.58	1.40	13.16	1.56	0.11
ITO+Cos-CbzNaph	40.94	9.92	38.16	2.55	13.21	1.22	0.19
ITO+IPA-MeO-2PACz	36.41	2.71	42.69	1.68	15.14	1.37	0.11
ITO+Cos-MeO-2PACz	39.69	2.92	40.43	2.07	13.70	1.19	0.15
ITO+IPA-2PACz	45.74	2.73	34.96	1.73	13.53	1.22	0.13
ITO+Cos-2PACz	52.77	3.08	30.88	2.12	10.20	0.95	0.21

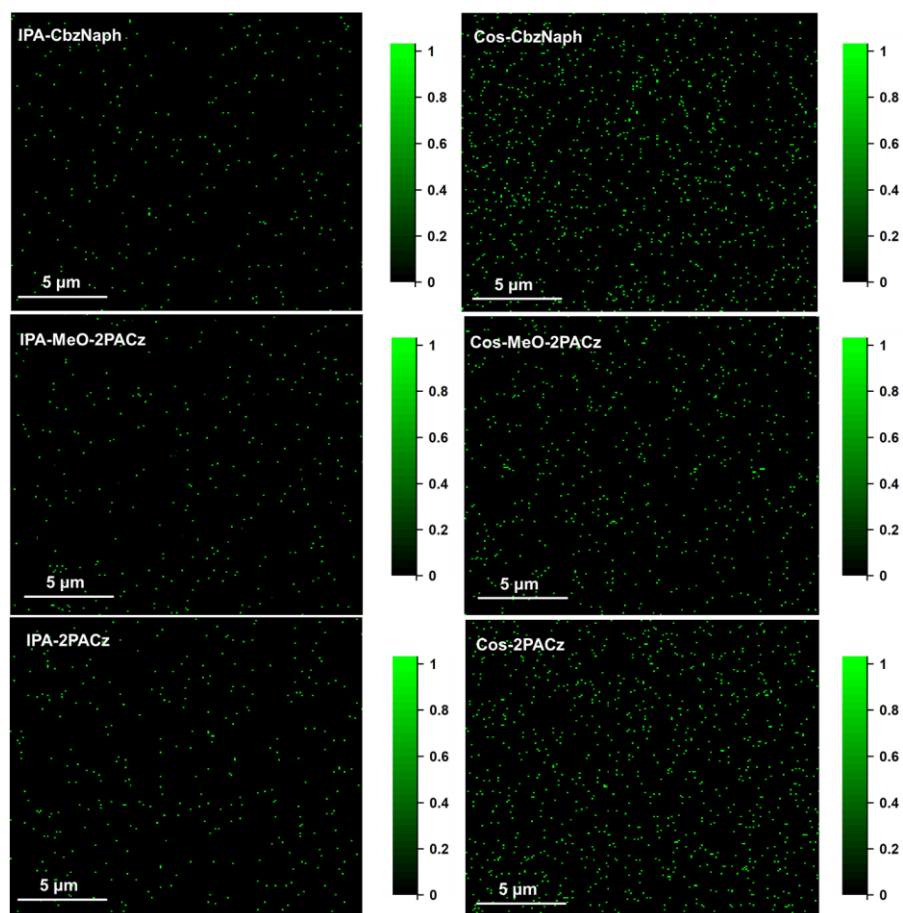


Figure S11. Elemental mapping of P for IPA-CbzNaph, Cos-CbzNaph, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-2PACz and Cos-2PACz modified ITO.

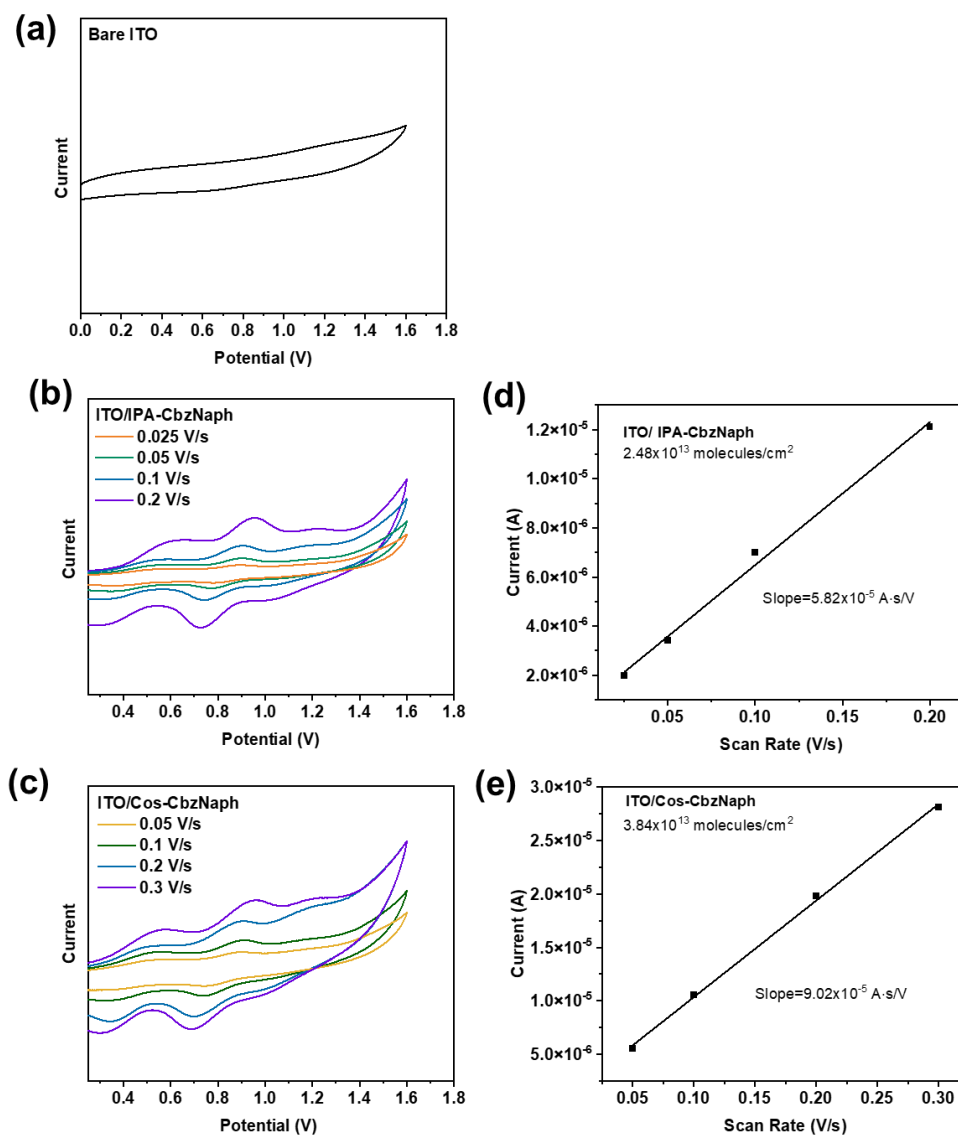


Figure S12. Cyclic voltammograms of (a) bare ITO; IPA-CbzNaph- and Cos-CbzNaph-modified ITO substrates measured in *o*-DCB solution under different scan rates (b-c) and their corresponding peak current vs. scan rate chart (d-e).

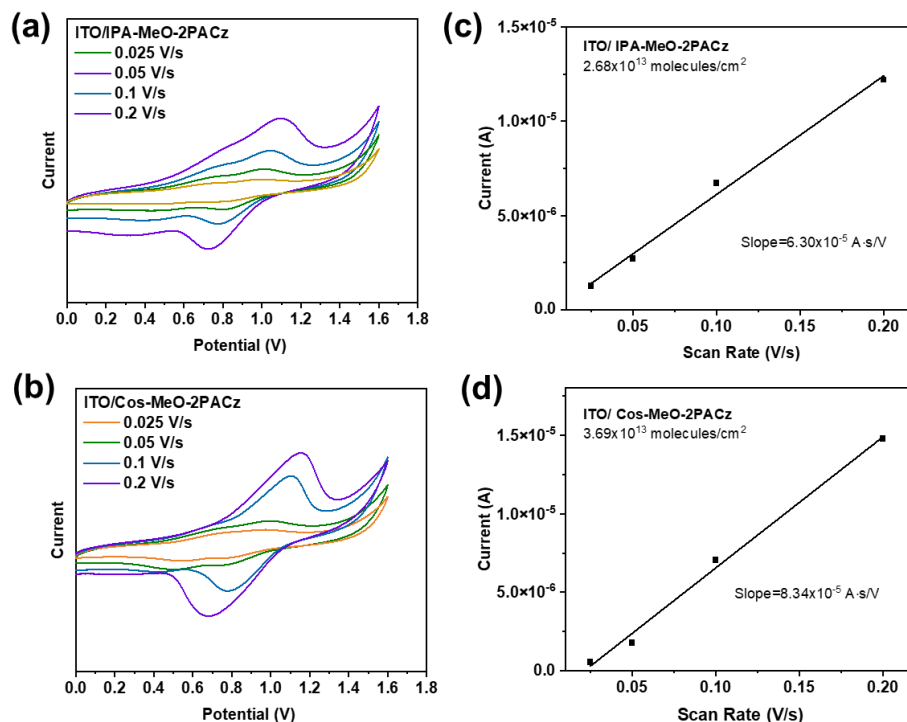


Figure S13. Cyclic voltammograms of IPA-MeO-2PACz and Cos-MeO-2PACz-modified ITO substrates measured in o-DCB solution under different scan rates (a-b) and their corresponding peak current vs scan rate chart (c-d).

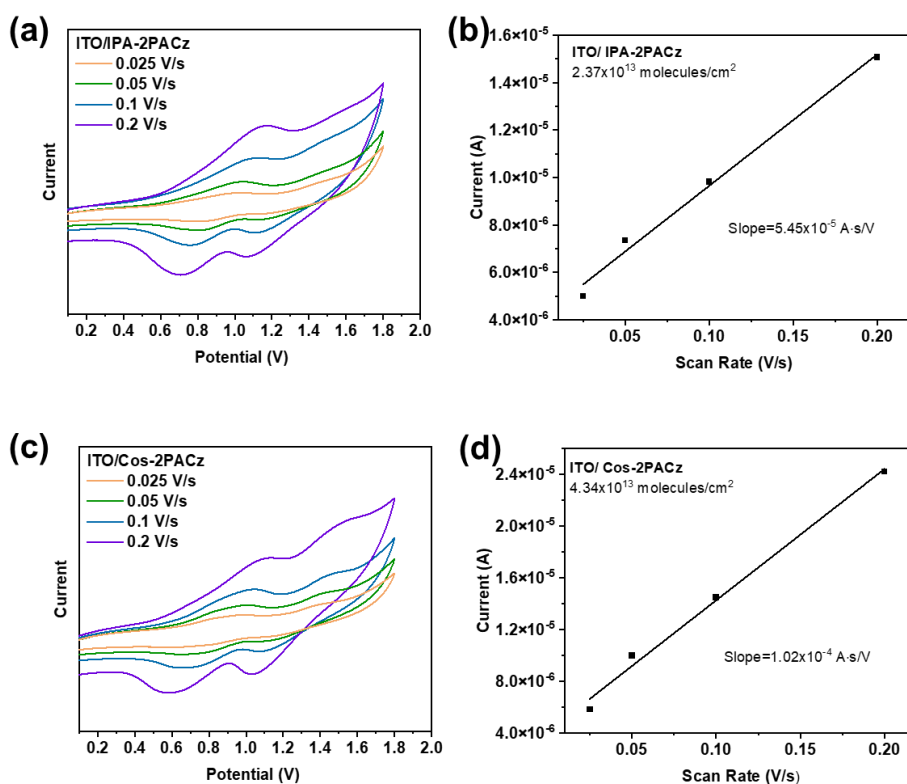


Figure S14. Cyclic voltammograms of the IPA-2PACz and Cos-2PACz-modified ITO substrates measured in o-DCB solution under different scan rates (a-b) and their corresponding peak current vs scan rate chart (c-d).

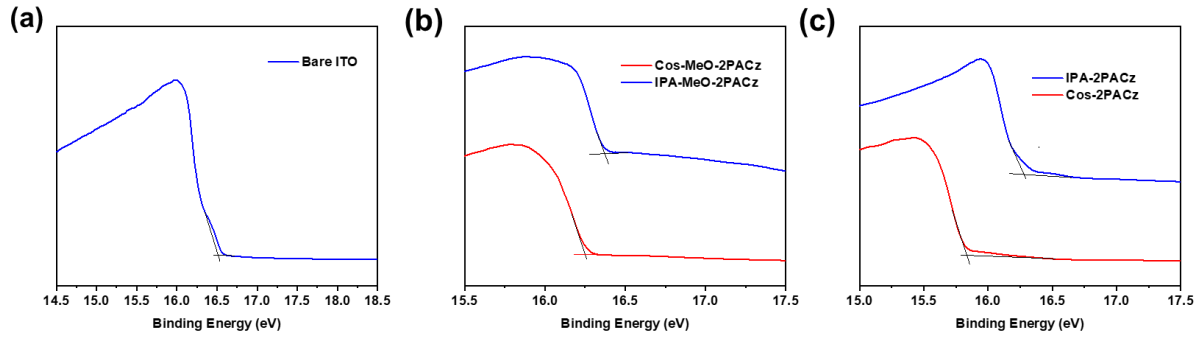


Figure S15. UPS spectra (using the He I lamp with a photon energy of 21.22 eV) of bare ITO, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-2PACz and Cos-2PACz modified ITO.

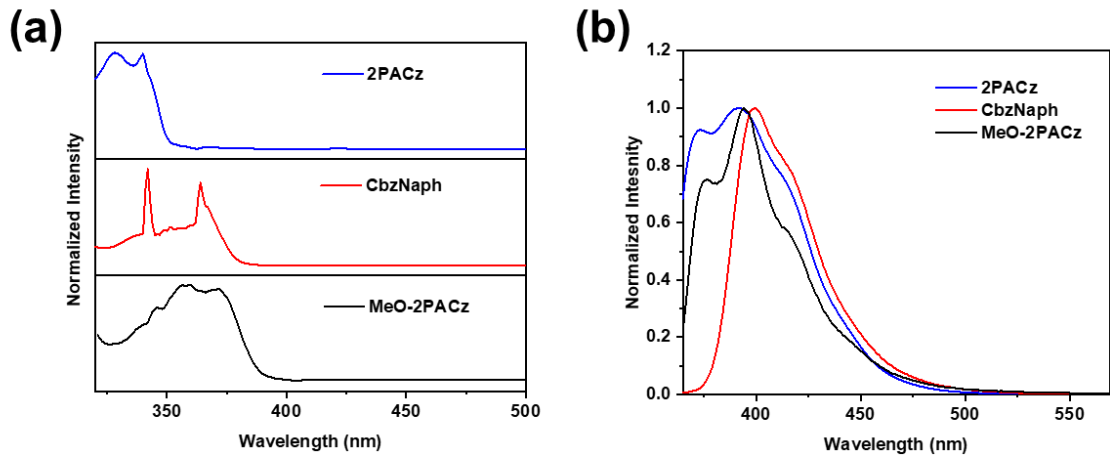


Figure S16. (a) UV-vis absorption spectra and (b) fluorescence emission spectra with an excitation wavelength at 340 nm for 2PACz, MeO-2PACz and CbzNaph in IPA.

Table S3. Lifetimes and weight fractions derived from time-resolved PL decay traces^[a].

Sample	τ_1 [ns]	Rel ₁ (%)	τ_2 [ns]	Rel ₂ (%)	τ_{avg} [ns]
ITO/IPA-CbzNaph	64.5	0.81	1909.8	99.19	1894.8
ITO/Cos-CbzNaph	36.6	1.04	1312.2	98.96	1298.9

^[a]The TRPL traces (Figure 5b) are fitted by bi-exponential fitting method with the equation $y(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) + y_0$. The average lifetime can be calculated with the equation $\tau_{avg} = (A_1 \tau_1 + A_2 \tau_2) / (A_1 + A_2)$.

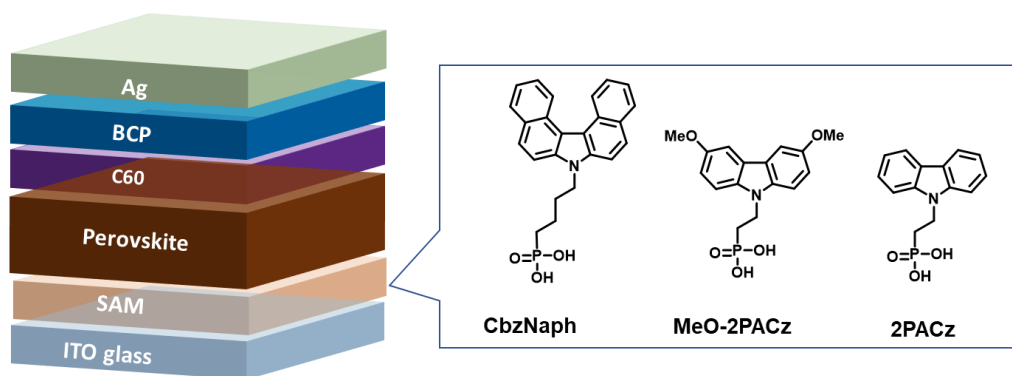


Figure S17. Device architecture of the inverted PSC and the chemical structures of SAMs used in this work.

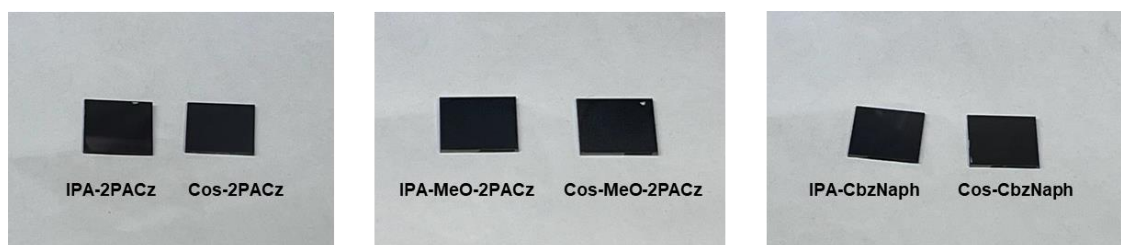


Figure S18. Photos of perovskite films deposited on different SAM substrates.

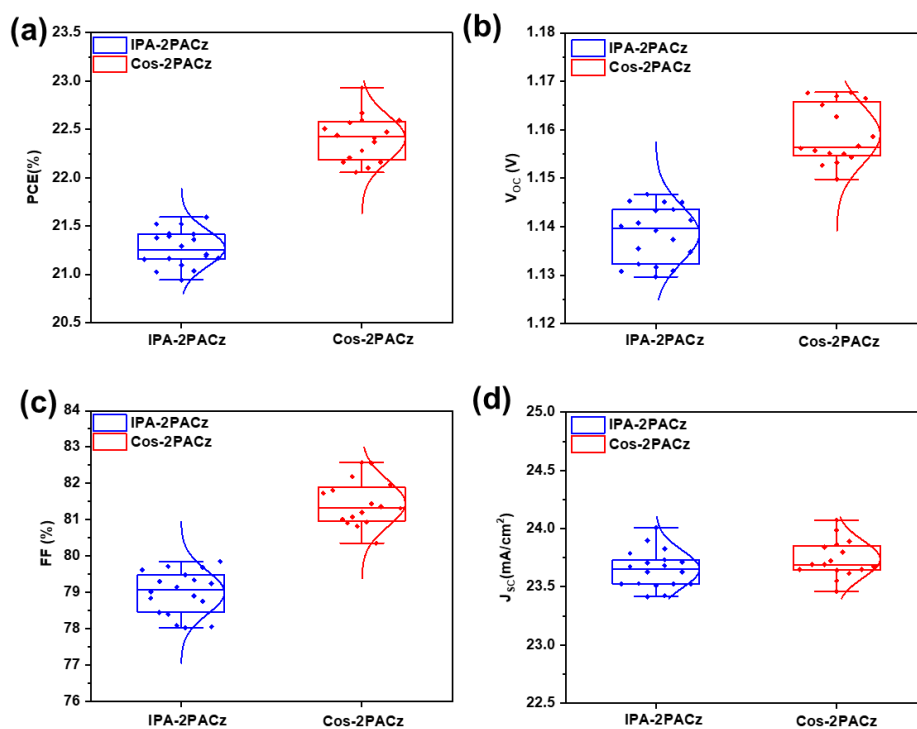


Figure S19. Box plots of (a) PCE, (b) V_{OC} , (c) FF and (d) J_{SC} of the PSCs employing SAM-modified substrates of IPA-2PACz and Cos-2PACz.

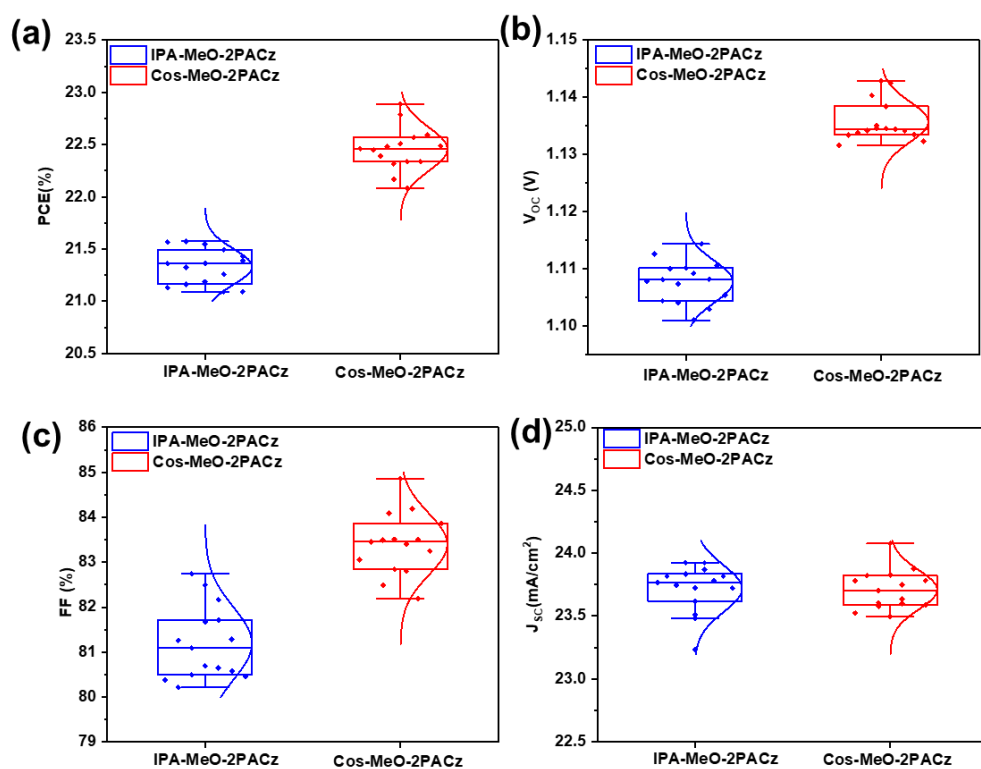


Figure S20. Box plots of (a) PCE, (b) V_{OC} , (c) FF and (d) J_{SC} of the PSCs employing SAM-modified substrates of IPA-MeO-2PACz and Cos-MeO-2PACz.

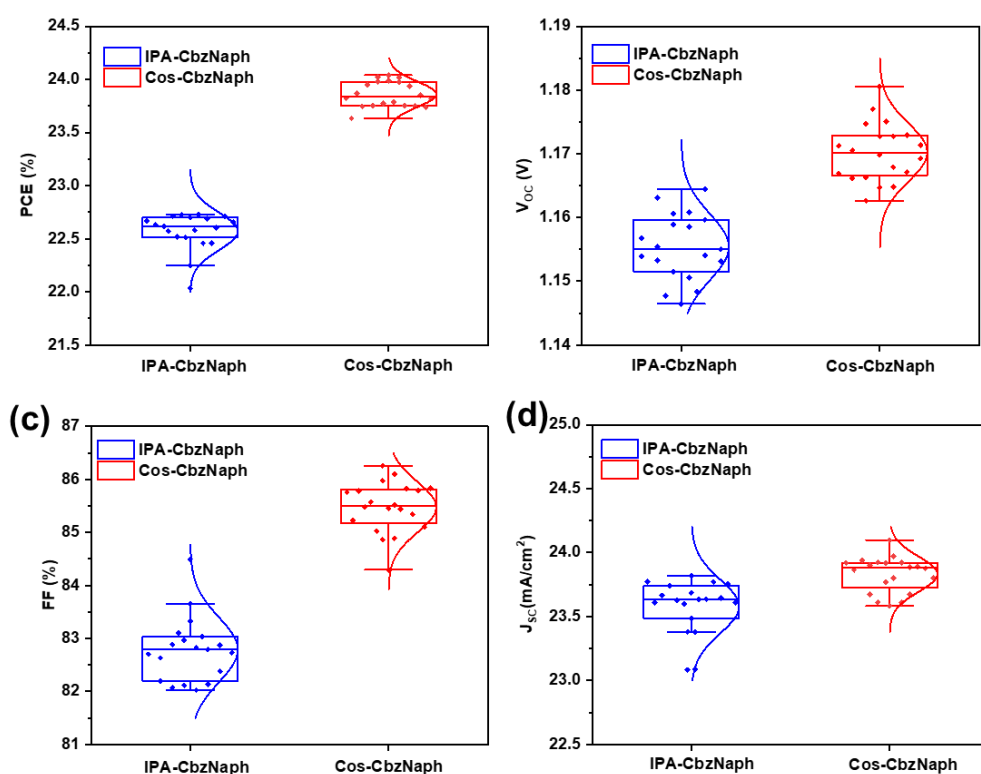


Figure S21. Box plots of (a) PCE, (b) V_{OC} , (c) FF and (d) J_{SC} of the PSCs employing SAM-modified substrates of IPA-CbzNaph and Cos-CbzNaph.

Table S4. Summarized photovoltaic parameters obtained from over 15 devices based on different SAM HSLs.

	V_{OC} (V)	FF (%)	J_{SC} (mA/cm ²)	PCE (%)
IPA-2PACz	1.135	79.61	23.90	21.59
	1.139±0.006	79.05±0.60	23.65±0.16	21.28±0.19
Cos-2PACz	1.167	82.58	23.80	22.93
	1.159±0.006	81.38±0.58	23.74±0.17	22.40±0.24
IPA-MeO-2PACz	1.108	81.68	23.84	21.57
	1.108±0.003	81.10±0.81	23.72±0.19	21.31±0.17
Cos-MeO-2PACz	1.134	84.86	23.78	22.89
	1.136±0.004	83.40±0.69	23.71±0.16	22.46±0.21
IPA-CbzNaph	1.152	83.32	23.69	22.73
	1.155±0.005	82.79±0.61	23.58±0.21	22.57±0.18
Cos-CbzNaph	1.170	86.26	23.80	24.02
	1.170±0.005	85.48±0.48	23.83±0.14	23.86±0.12

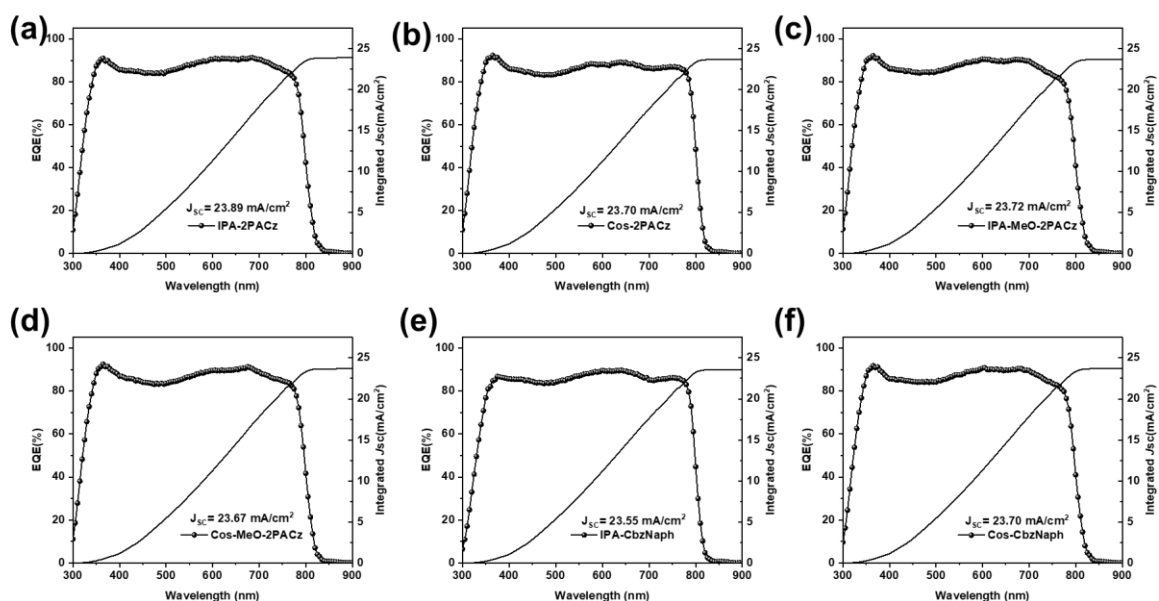


Figure S22. EQE of the champion device employing IPA-2PACz, Cos-2PACz, IPA-MeO-2PACz, Cos-MeO-2PACz, IPA-CbzNaph and Cos-CbzNaph HSL.

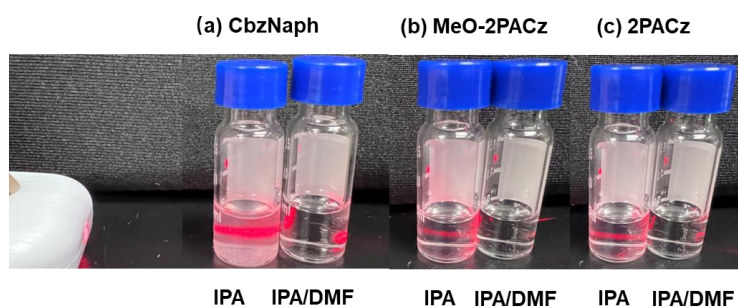


Figure S23. The images of (a) CbzNaph (b) MeO-2PACz and (c) 2PACz solution prepared by co-solvent (left) and IPA (right) after 30 days' storage.

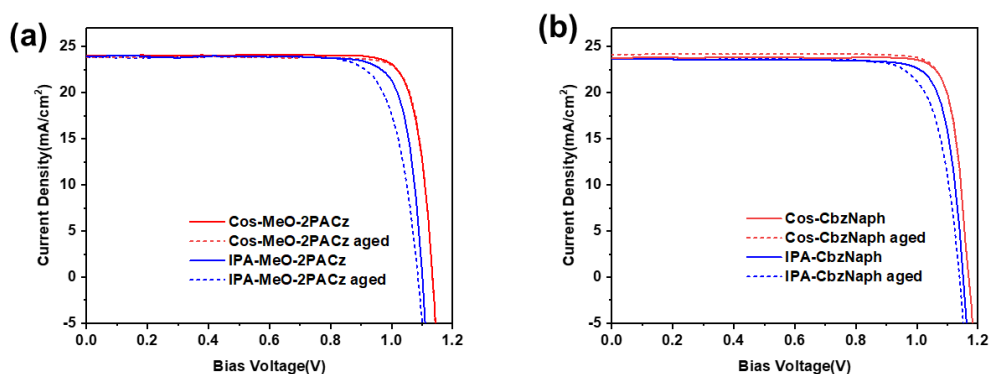


Figure S24. J-V curves of PSC devices based on different SAM inks before and after aging.

Table S5. Summarized photovoltaic parameters obtained from over 15 devices based on HSLs processed from SAM inks after 30 days storage.

	V_{OC} (V)	FF (%)	J_{SC} (mA/cm ²)	PCE (%)
Aged	1.086	79.39	23.81	20.53
IPA-MeO-2PACz	1.090±0.011	77.49±1.36	23.60±0.32	19.95±0.45
Aged	1.134	85.03	23.86	23.00
Cos-MeO-2PACz	1.133±0.005	84.13±0.74	23.82±0.21	22.70±0.22
Aged	1.140	79.59	23.75	21.55
IPA-CbzNaph	1.131±0.010	80.60±1.74	23.02±0.57	20.99±0.32
Aged	1.170	85.65	24.14	24.19
Cos-CbzNaph	1.170±0.003	85.36±0.32	23.87±0.29	23.87±0.29

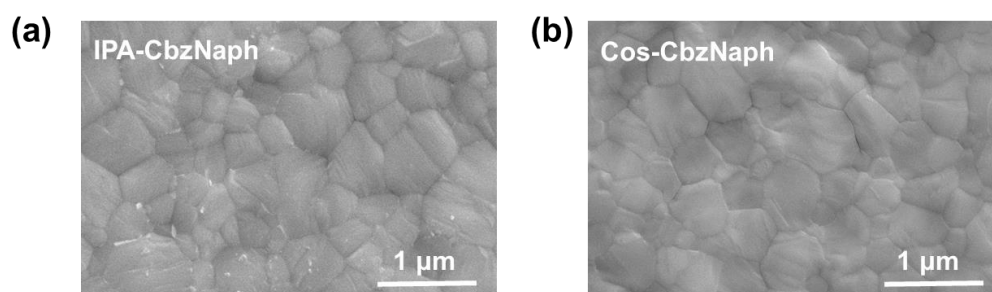


Figure S25. SEM images of perovskite film based on IPA-CbzNaph and Cos-CbzNaph modified ITO.

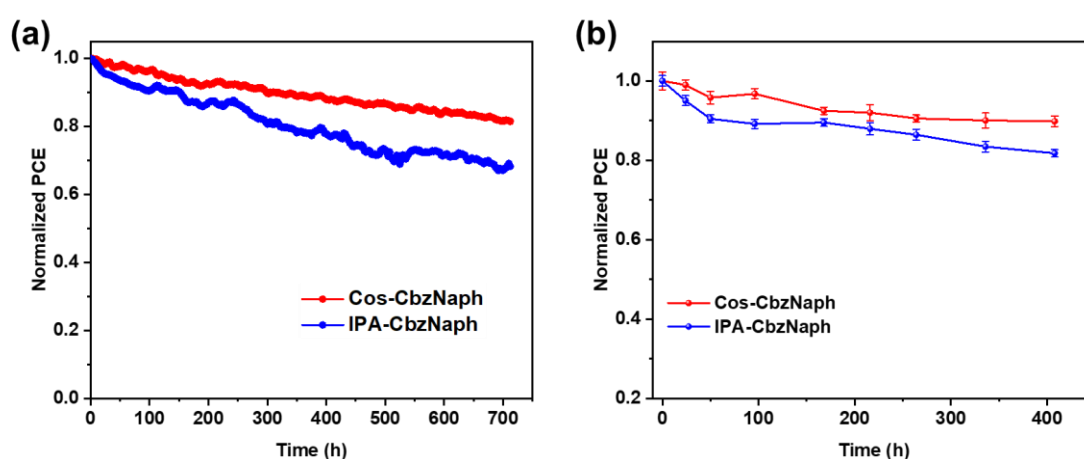


Figure S26. (a) The normalized PCE is shown as a function of time under one-sun continuous illumination, MPP tracking, unencapsulated, N₂ atmosphere and operating cell temperature of ~45 °C. (b) The normalized PCE of PSC devices aged at 65 °C in nitrogen atmosphere, the error bar indicates the standard deviation of 16 devices.